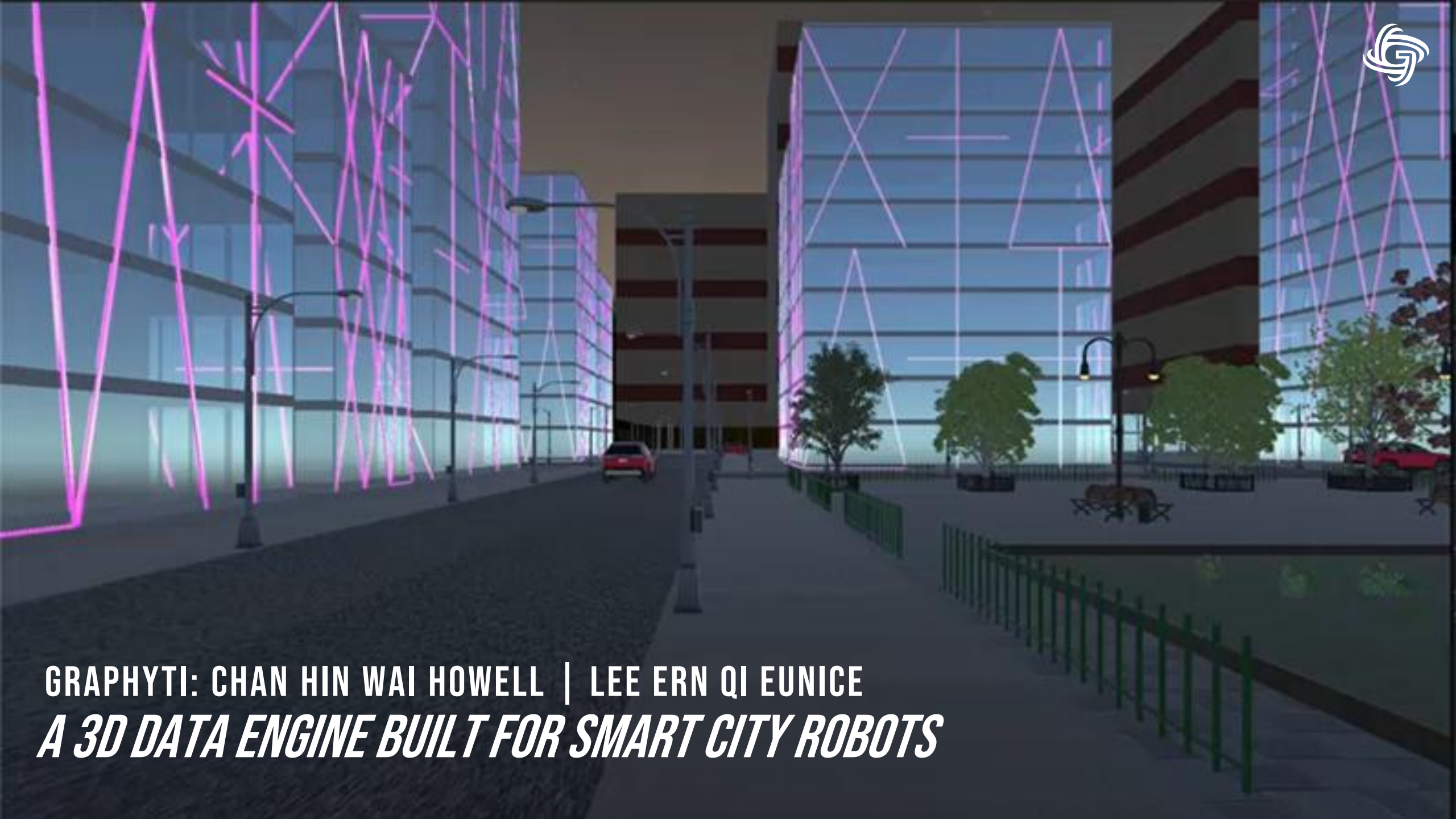


GRAPHYTI: CHAN HIN WAI HOWELL | LEE ERN QI EUNICE

A 3D DATA ENGINE BUILT FOR SMART CITY ROBOTS

A MAJOR BOTTLENECK IN ROBOTICS IS FINDING THE "PERFECT DATA" TO TRAIN YOUR ROBOTS

Training Data must differ in POV

Training Data must differ in Use Cases

Therefore, this makes finding the "perfect data" extremely difficult

CURRENTLY, COMPANIES COLLECT DATA THROUGH MANUAL COLLECTION

Companies hire **data collectors** who use **expensive camera hardware** to capture images or videos across **diverse environments** and conditions

Expensive

~US$650k

Data collection cost includes hardware, annotation, travel logistics and manual labour

Long Process

>6 Months

Data collection takes at least 3 months and up to 1 year for top AI models

Limited High-Quality Copyright-Free Data

~5%

Data bottlenecks are hit quickly without proprietary data as less than 5% of online data are suited for training

Therefore, Robotics Companies need an **affordable** and **efficient** way to get customized datasets

SOLUTION: GRAPHYTI, A 3D SYNTHETIC DATA PLATFORM

A 3D Data Engine Built for Training Robots

A platform for companies to create datasets using text prompts

---

Affordable

~80%

cost savings

3D environmental capture eliminates manual real-world data collection costs

Simple & Fast

<5min

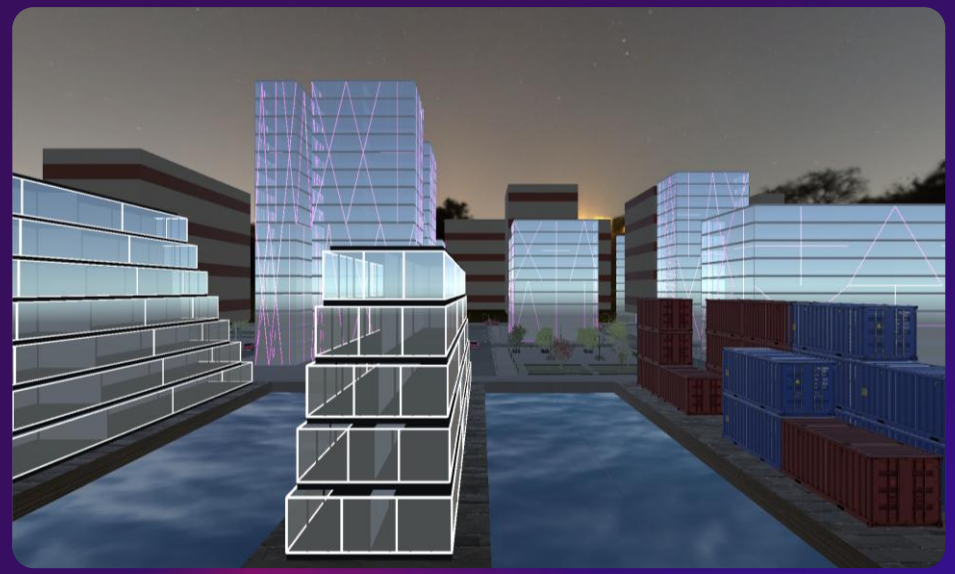
Using a No-Code UI, creating a full dataset for 1 scene takes under 5min

Diverse, High-Quality Copyright-Free Data

∞

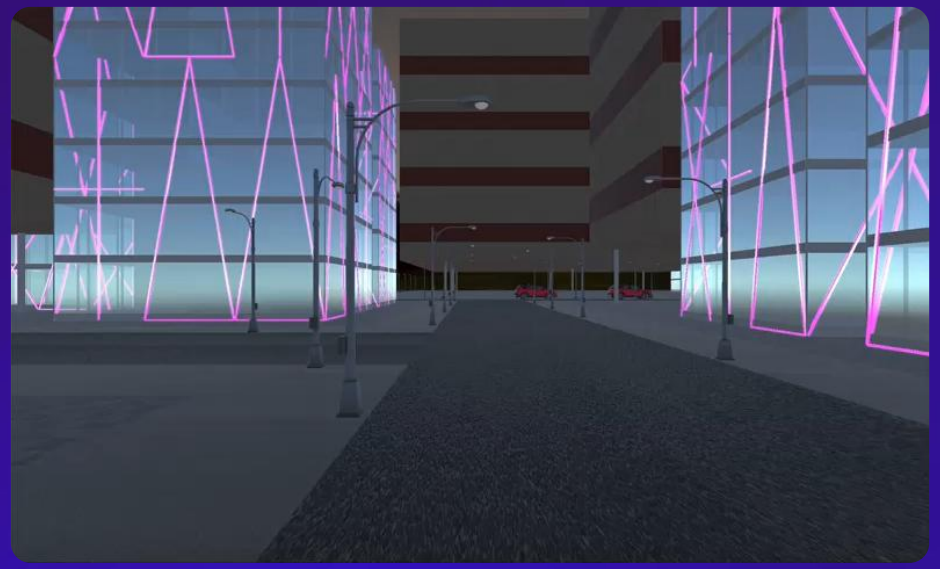
Infinite unique 3D worlds designed for training Visual AI

CORE: ALGORITHM-BASED 3D SCENE GENERATION



Minimal AI Approach

- Not using AI images to train AI
- True 3D Geometry



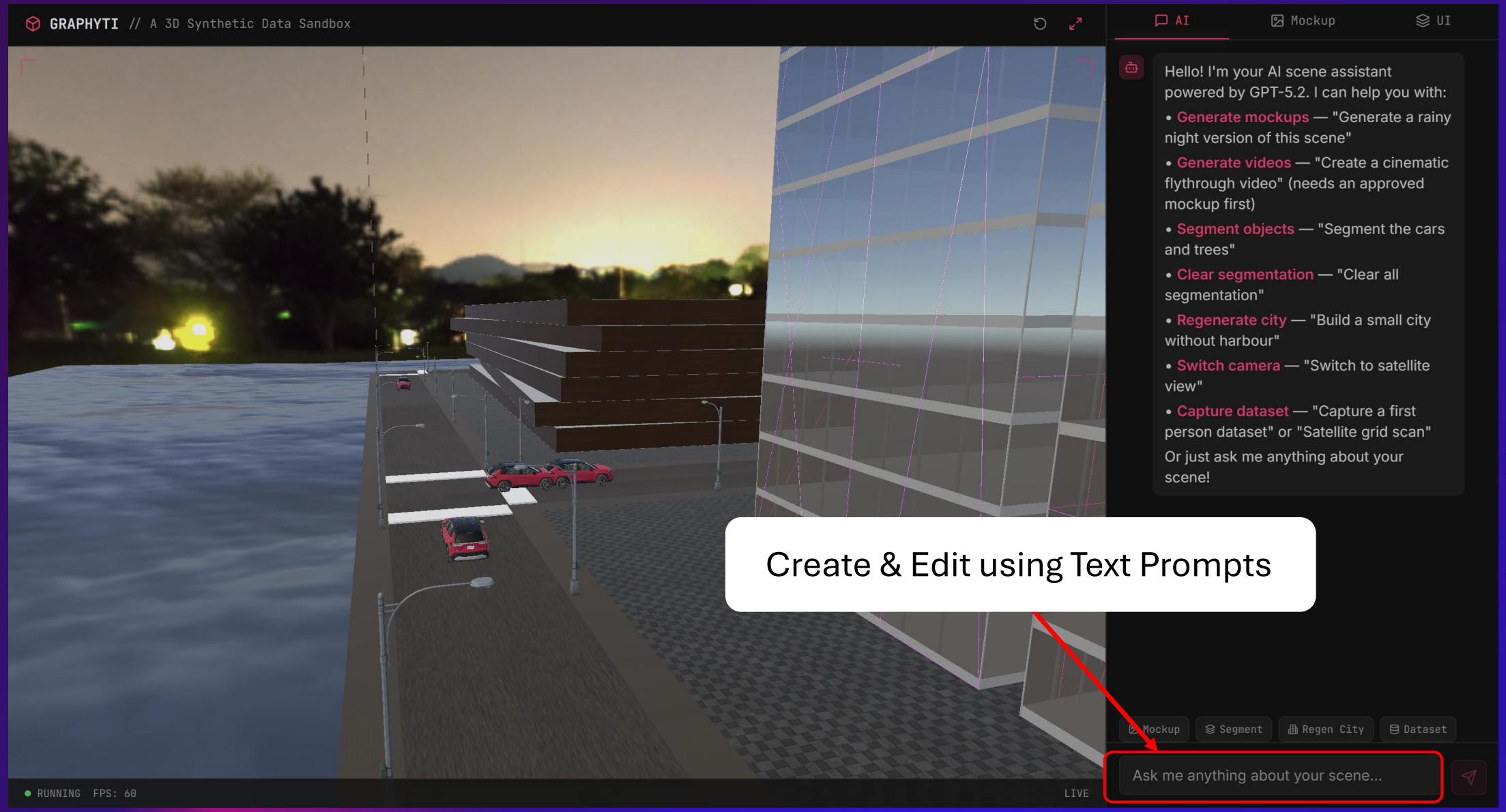
Procedural/Algorithmic Approach

- Layouts, Assets controlled by Algorithms
- Reproducible & Controllable → No Black Box

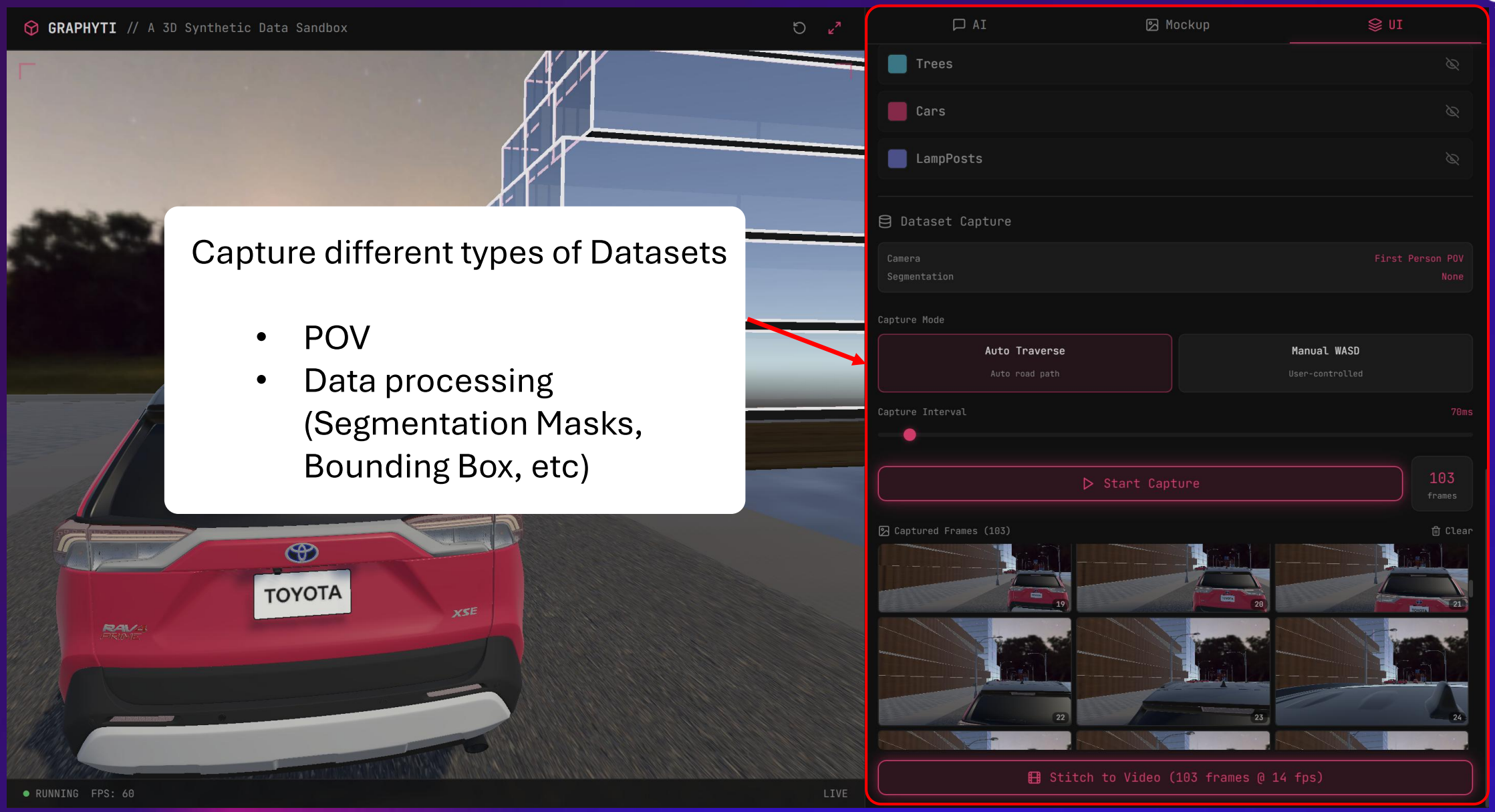
AI as a Design Co-Pilot, Not a Creator

- LLM to interpret natural language prompts
- Translate into algorithmic parameters for 3D scene generation

FEATURES: AI DESIGN CO-PILOT



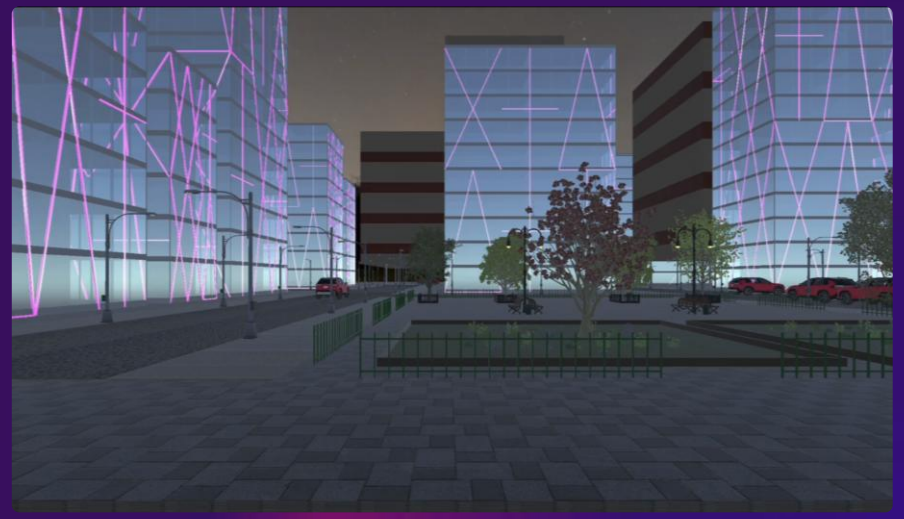
FEATURES: AUTO DATASET CAPTURE



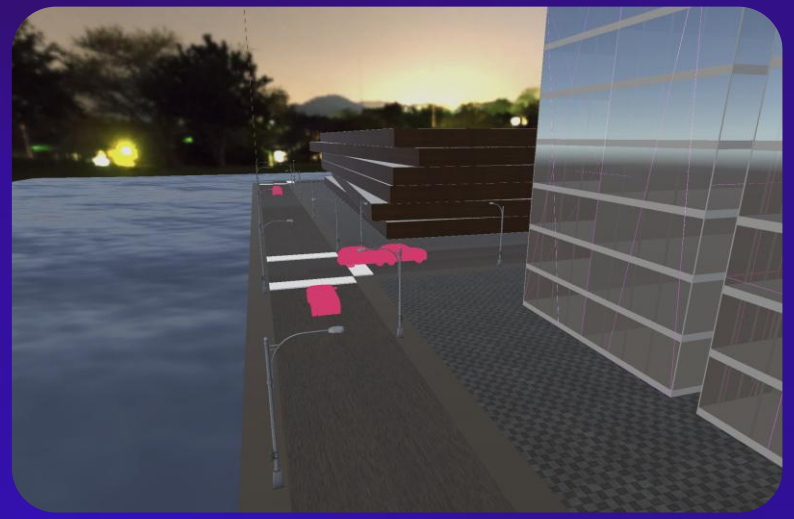
Capture different types of Datasets

- POV
- Data processing (Segmentation Masks, Bounding Box, etc)

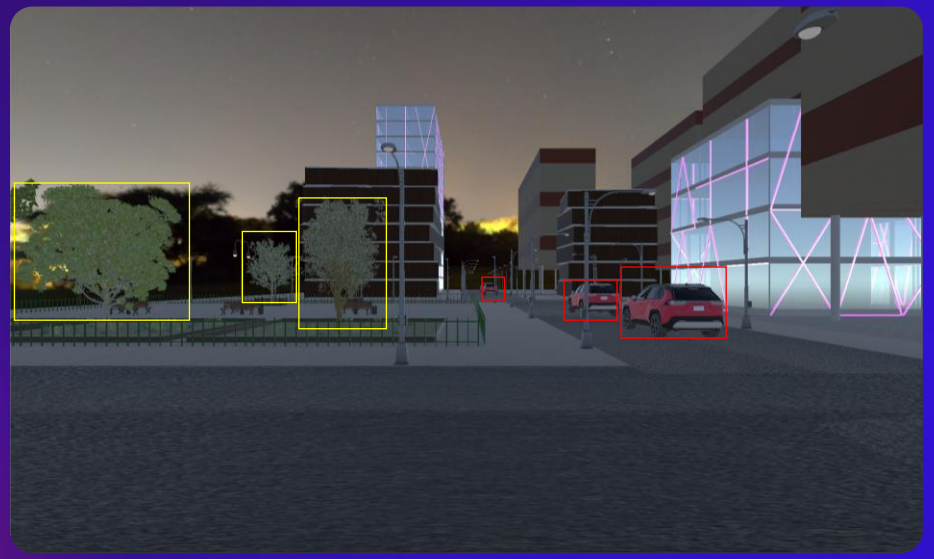
FEATURES: DIFFERENT DATA PROCESSING OUTPUTS



Raw RGB Image/Video



Segmentation Masks



Bounding Boxes

etc...

OUR VISION FOR GRAPHYTI

GRAPHYTI WILL EMPOWER TEAMS OF ANY SIZE & ABILITY TO CREATE PROPRIETARY 3D DATASETS

USE CASES

3D Datasets for Robotics

Robotics

Autonomous Vehicles

Military Drones

World Models

Digital Twins

POTENTIAL USERS

Boston Dynamics

Venti

DSO National Laboratories

HTX

World Labs

Nanyang Technological University Singapore

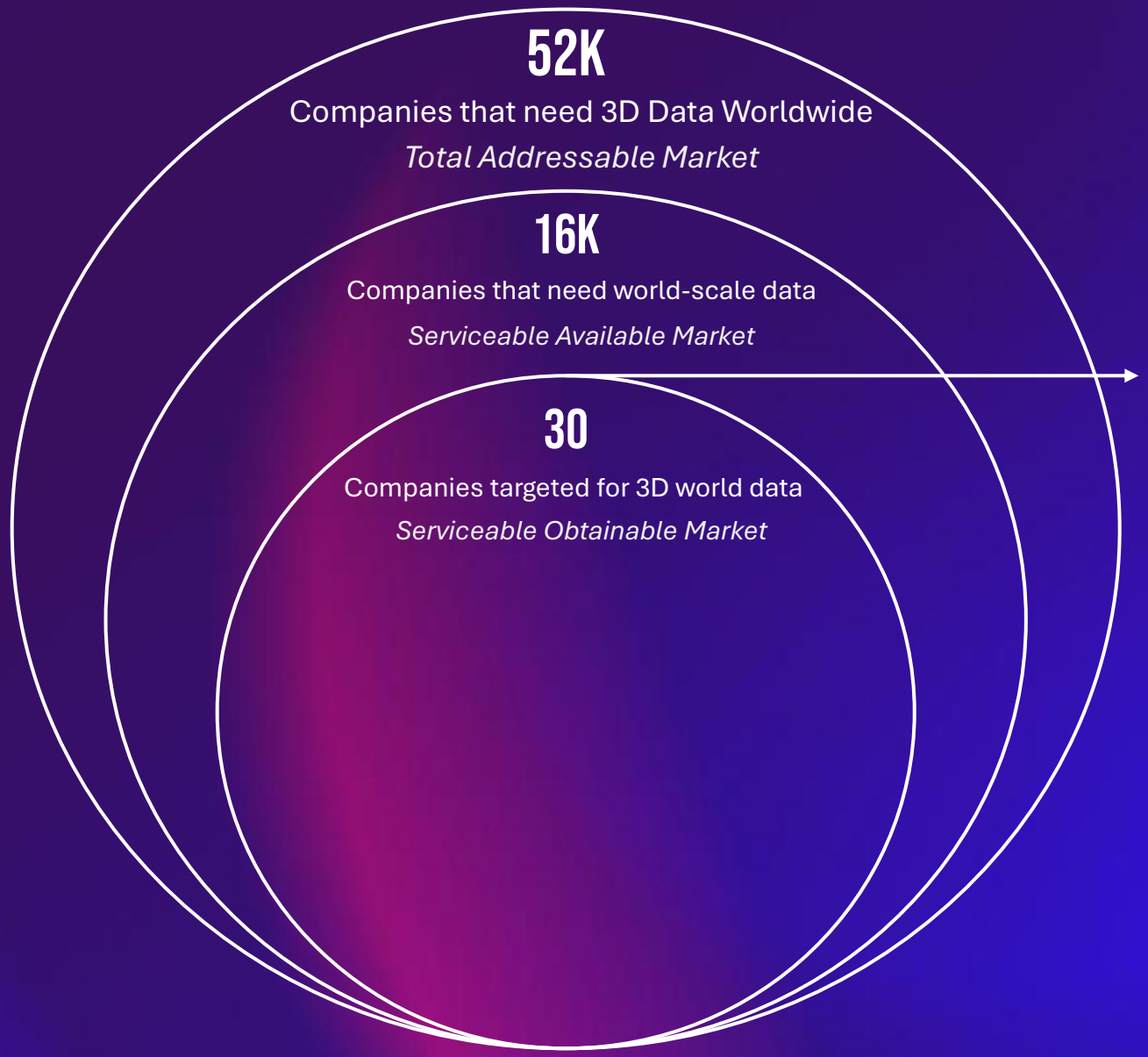
NUS National University of Singapore

GovTech Singapore

Augmentus Robotics Simplified

Eureka Robotics

MARKET SIZE DEEP DIVE



B2B PRICING TIERS

20% High-Budget Tier	240K/year		20k/month
50% Mid-Budget Tier	120K/year	*or*	10k/month
30% Low-Budget Tier	24K/year		2k/month

Projected ARR: US$3.46M by 2028

COMPETITIVE ADVANTAGE

We're building the world's largest suite of procedural generation algorithms usable through a promptable interface

Graphyti		Speed & Scalability	Cost
3 engineers + 2 artists	1 week →	Full-scale 3D world with infinite variations (layouts, Style, etc.)	2k/month for small teams | 10k/month for large teams

Competitors

6 engineers + 3 artists	1 month →	3D world with limited variations & layouts	2.5k/month for small teams | 15k/month for large teams

First to build 3D data for Multiple Use Cases across Multiple Industries

Compatible data formats for all visual AI models for AI/Graphics Research, Robotics, Entertainment/Advertising

NETFLIX
World Labs
HTX
Disney
Google DeepMind
BostonDynamics

OUR TEAM

Howell Chan

Cofounder | AI/Unity Developer

- B.Eng in CS + M.Sc. In Tech Management@ NTU
- Work experience in AI & full-stack engineering
- Former NTU researcher in Computer Graphics

Eunice Lee

Cofounder | Graphics/Unity Developer

- B.Eng in CS + M.Sc. In Tech Management@ NTU
- Work experience in Computer Graphics
- Former NTU researcher in Computer Graphics

We have won multiple prestigious competitions while doing this as a side project.

Various companies have expressed interests in working with us.

We are graduating at the end of 2026, and we'll be ready to take Graphyti full-time